Oluwatosin Ayeni

Staff Software Engineer, Inference — Anthropic (London)

Glasgow, Scotland, UK · 07727305230 · ayenioluwatosinolawale@gmail.com
github.com/AyeniOluwatosinOlawale · linkedin.com/in/oluwatosinolawaleayeni

PROFILE

Staff-calibre AI Inference Engineer with proven expertise in distributed LLM serving, GPU infrastructure, and production ML systems at scale. Benchmarked vLLM and SGLang on H100 NVL — 5,965 tok/s peak throughput, 1,443× TTFT speedup via prefix caching, 100% SLO compliance at concurrency 128. Built production LLM inference platform on GKE (NVIDIA L4, 723 requests, 1.25M+ tokens) and Inference Gateways delivering 14× throughput and 3.87× latency improvement. Deep expertise in batching, KV cache optimisation, routing, and Kubernetes GPU infrastructure. Ready to relocate to London.

CORE COMPETENCIES

Distributed Inference	Tensor parallelism (TP=2), pipeline parallelism, KV cache-aware routing, multi-node GPU orchestration, batching strategies (continuous, dynamic, chunked prefill)
Inference Engines	vLLM (PagedAttention, continuous batching), SGLang (RadixAttention, prefix caching), TensorRT-LLM, LiteLLM Proxy — production deployment and benchmarking
ML Infrastructure	Kubernetes (HPA/KEDA), GKE, Docker, Prometheus/Grafana, Argo CD/Rollouts, Terraform, Traefik, Redis, PostgreSQL, GitHub Actions
GPU & CUDA	NVIDIA Nsight Systems/Compute, memory bandwidth profiling, DCGM Exporter, BFloat16/ AWQ quantisation, NVIDIA L4/H100 NVL production experience
LLM Systems	Speculative decoding, multi-provider routing, A/B weighted round-robin, Redis response caching, KEDA autoscaling on GPU queue depth, output scrubbing

PROFESSIONAL EXPERIENCE

Jan 2025–present — **Founder & Lead Engineer** — *Nexora Academic*, Glasgow, UK

- H100 NVL benchmark: vLLM v0.30 vs SGLang v0.5.20 — Qwen3-8B BF16, TP=2. Peak: 5,965 tok/s; RadixAttention: 1,443× TTFT speedup (39 ms vs 55,565 ms) at 131K-token context; 100% SLO compliance up to concurrency 128. Live: inference-web-theta.vercel.app
- Built *inference-optimizer*: production AI Inference Gateway — 3.87× faster (parallel execution), 61.4% cost reduction (prompt caching), 14× throughput (concurrency), 64.3% cheaper (model routing). 13 optimisations, 4 providers.
- Built *vllm-k8s-stack*: Kubernetes LLM serving — KV-cache-aware routing, HPA autoscaling on GPU pools, Prometheus/Grafana observability.
- Built *inference-harness*: unified benchmark framework spanning closed APIs + vLLM/SGLang/TensorRT-LLM with GPU profiling and live dashboard.
- Designed 13-phase AI Inference Engineering curriculum covering GPU architecture, CUDA optimisation, serving frameworks, and distributed inference.

Jan 2023–present — **Founder & AI Consultant** — *Vertinab / AnaburyAI Technology*, Glasgow, UK

- 28× LLM cost reduction deploying Qwen2.5-7B on RTX 4090 via vLLM/SGLang with continuous batching and KV cache optimisation.
- CUDA KV cache kernel profiling with NVIDIA Nsight: identified memory bandwidth bottleneck; 3.8× throughput improvement; 18× latency reduction via continuous batching redesign.
- Fine-tuned Mistral 7B with LoRA/PEFT on radiology corpus: +22% diagnostic accuracy, −35% hallucination rate; deployed as production clinical inference API.

2024–present — **Senior Software Engineer (Contract)** — *EduTechConnect / Stellar Labs*, Remote

- Real-time NLP pipeline (WhatsApp → Zoho CRM → Supabase) with sub-500 ms end-to-end latency under concurrent load.

Mar 2026–present **AI Automation Engineer** — *EduTech Global*, Remote

- LLM-backed student chatbots with structured output extraction and multi-step reasoning for Ahmadu Bello and Babcock universities.

2018–2022 **Data Analyst** — *Bytelab Technologies Ltd*, Nigeria

KEY PROJECTS

llm-inference-platform (GKE) Production on NVIDIA L4, vLLM + SGLang, LiteLLM gateway, KEDA autoscaling, 27-target Prometheus stack, Argo CD GitOps, Traefik TLS. LIVE: 723 req, 1.25M+ tokens. GitHub

llm-inference-benchmark (H100) 2× H100 NVL 94 GB, CUDA 13.4. vLLM v0.30 vs SGLang v0.5.20, Qwen3-8B BF16, TP=2. RadixAttention: 1,443× TTFT speedup. Peak: 5,965 tok/s. SLO 100% at concurrency 128. Live results

inference-optimizer Production Inference Gateway: 13 optimisations, multi-provider (Claude, GPT, Gemini), 3.87× faster, 14× throughput, 61.4% cost reduction. GitHub

vllm-k8s-stack Kubernetes LLM serving: vLLM, KV-cache-aware routing, HPA autoscaling, Prometheus/Grafana. GitHub

open-model-engine GPU-first inference benchmarking: vLLM, SGLang, TensorRT-LLM, CUDA profiling, quantisation, speculative decoding. GitHub

EDUCATION

2022–2023 **MSc Artificial Intelligence and Applications, Distinction** — *University of Strathclyde*, Glasgow, UK

2010–2016 **MTech / BSc Physics** — *Federal University of Technology Akure*, Nigeria

OPEN SOURCE

GitHub 43 public repos — github.com/AyeniOluwatosinOlawale